LO7-DB Design using UML

Chapter 10 in Edition 6 and Chapter 3.8 and 4.6 in Edition 7

Exercise from previous lecture:

- Consider a GRADE_BOOK database in which instructors within an academic department record points earned by individual students in their classes. The data requirements are summarized as follows:
- Each student is identified by a unique identifier, first and last name, and an e-mail address.
- Each instructor teaches certain courses each term. Each course is identified by a course number, a section number, and the term in which it is taught. For each course he or she teaches, the instructor specifies the minimum number of points required in order to earn letter grades A, B, C, D, and F. For example,90 points for an A, 80 points for a B, 70 points for a C, and so forth.
- Students are enrolled in each course taught by the instructor.
- Each course has a number of grading components (such as midterm exam, final exam, project, and so forth). Each grading component has a maximum number of points (such as 100 or 50) and a weight (such as20% or 10%). The weights of all the grading components of a course usually total 100.
- Finally, the instructor records the points earned by each student in each of the grading components in each of the courses. For example, student *1234* earns 84 points for the midterm exam grading component of the section 2 course CSc2310 in the fall term of 2009. The midterm exam grading component may have been defined to have a maximum of 100 points and a weight of 20% of the course grade.
- Design an enhanced entity–relationship diagram for the grade book database..

Solution—Grade book database

1. List Entity types and their attributes.
2. List and describe relationship types and their cardinality ratios
3. Present EER

University Example - Requirements

- The database keeps track of three types of persons: employees, alumni, and students. A person can belong to one, two, or all three of these types. Each person has a name, SSN, sex, address, and birth date.
- Every employee has a salary, and there are three types of employees: faculty, staff, and student assistants. Each employee belongs to exactly one of these types. For each alumnus, a record of the degree or degrees that he or she earned at the university is kept, including the name of the degree, the year granted, and the major department. Each student has a major department.
- Each faculty has a rank, whereas each staff member has a staff position. Student assistants are classified further as either research assistants or teaching assistants, and the percent of time that they work is recorded in the database. Research assistants have their research project stored, whereas teaching assistants have the current course they work on.
- Students are further classified as either graduate or undergraduate, with the specific attributes degree program (M.S., Ph.D., M.B.A., and so on) for graduate students and class (freshman, sophomore, and so on) for undergraduates.

University Example - Notations

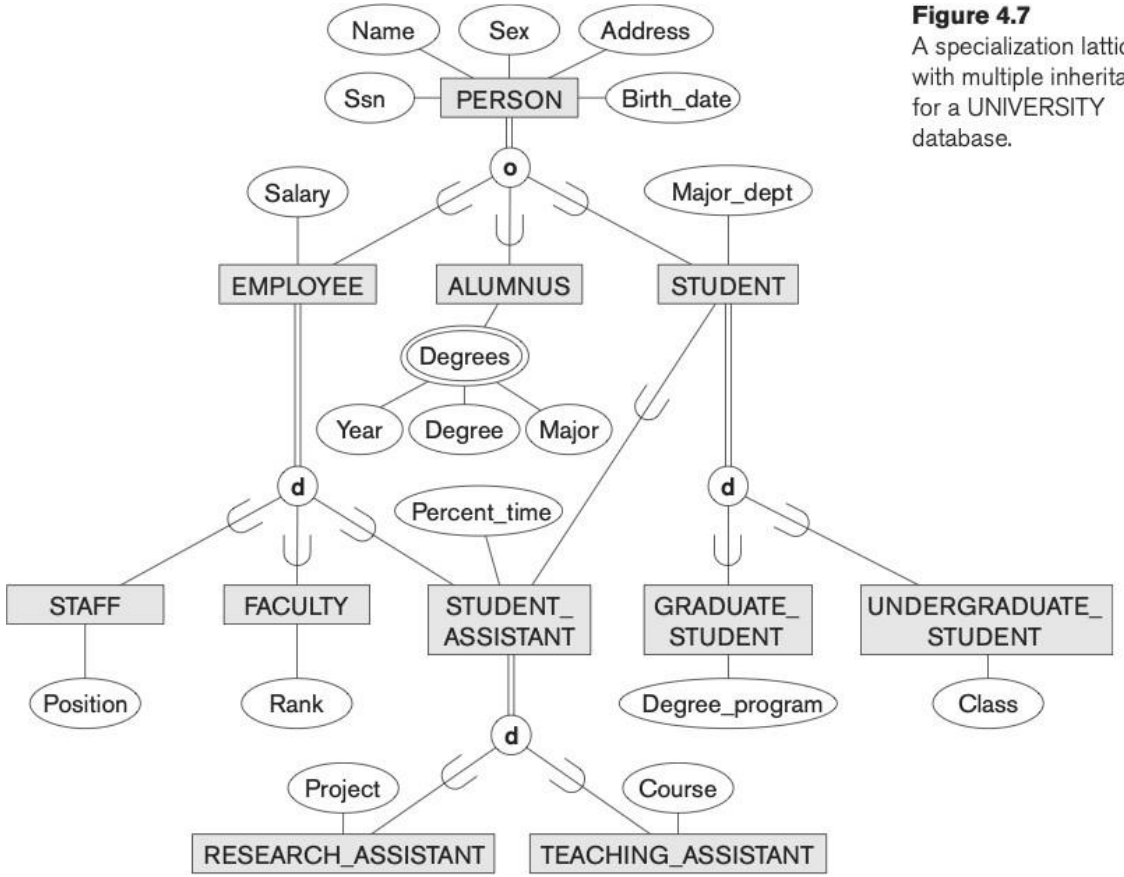

Figure 4.7
A specialization lattice with multiple inheritance for a UNIVERSITY database.

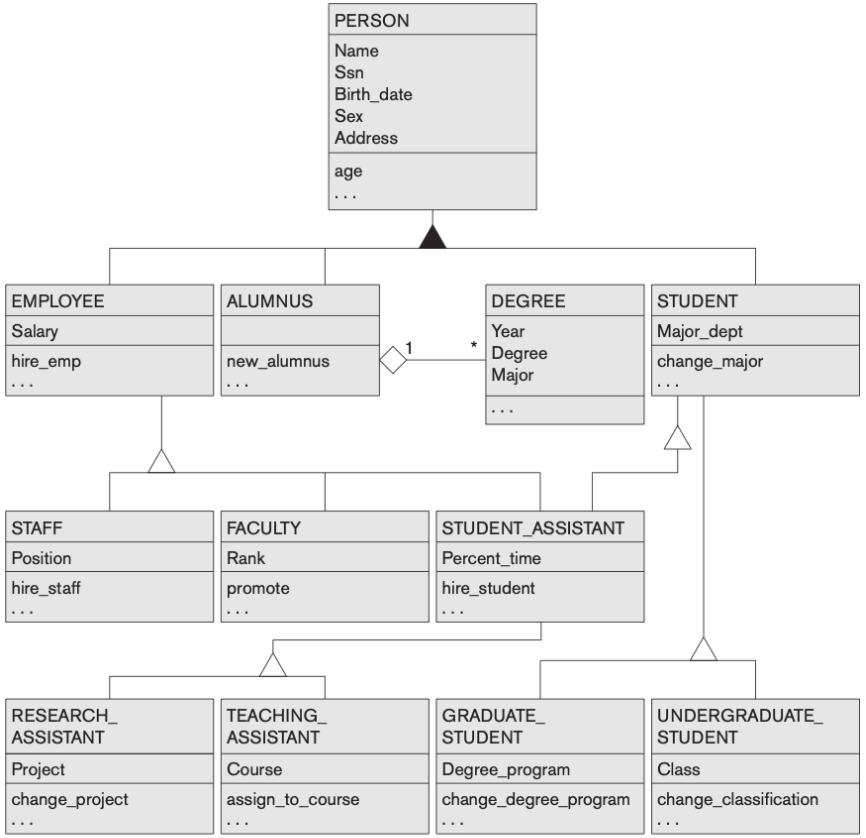

Figure 4.10
A UML class diagram corresponding to the EER diagram in Figure 4.7, illustrating UML notation for specialization/generalization.

Alternative diagrammatic notations

- Multiple methodologies exist to represent database design.
- ER/EER diagrams are a specific notation for dispalying the concepts of the model diagrammatically
- DB design tools use many alternative notations for the same or similar concepts
- One alternative notation uses UML class diagrams

Introduction to UML

- **Class-** comprises of attributes and operations. Attribute domains (data types) are optional. Operations are not specified on ER diagrams. In UML operations are class methods(behaviour); in DB implementations they may correspond to stored procedures.
- **Modelling complex attributes** - Composite attributes modelled as structured domains, e.g. Name. Multivalued attributes are modelled as a separate class, e.g. LOCATION.
- **Relationships** are modelled using associations.
  - Associations have multiplicity and roles
  - There are two variants of association, aggregation and composition.
  - Associations can be unidirectional (arrowed) or bidirectional.
  - Reflexive associations link to self.
  - Class ≈ entity type; association ≈ relationship; multiplicity ≈ (min,max)constraint.

UML notation used in data modelling

- **Association attributes**: Relationship attributes are shown in a box attached by a dashed line to the association (e.g., Start_date, Hours)e box.

- **Ordering (optional):** Related objects along an association may be specified as ordered (e.g., employees per department ordered by *Start_date*)

- **Weak entities:** Weak entities can be modelled using a **qualified association/aggregation**; the qualifier attribute is the **discriminator** (ER partial key)

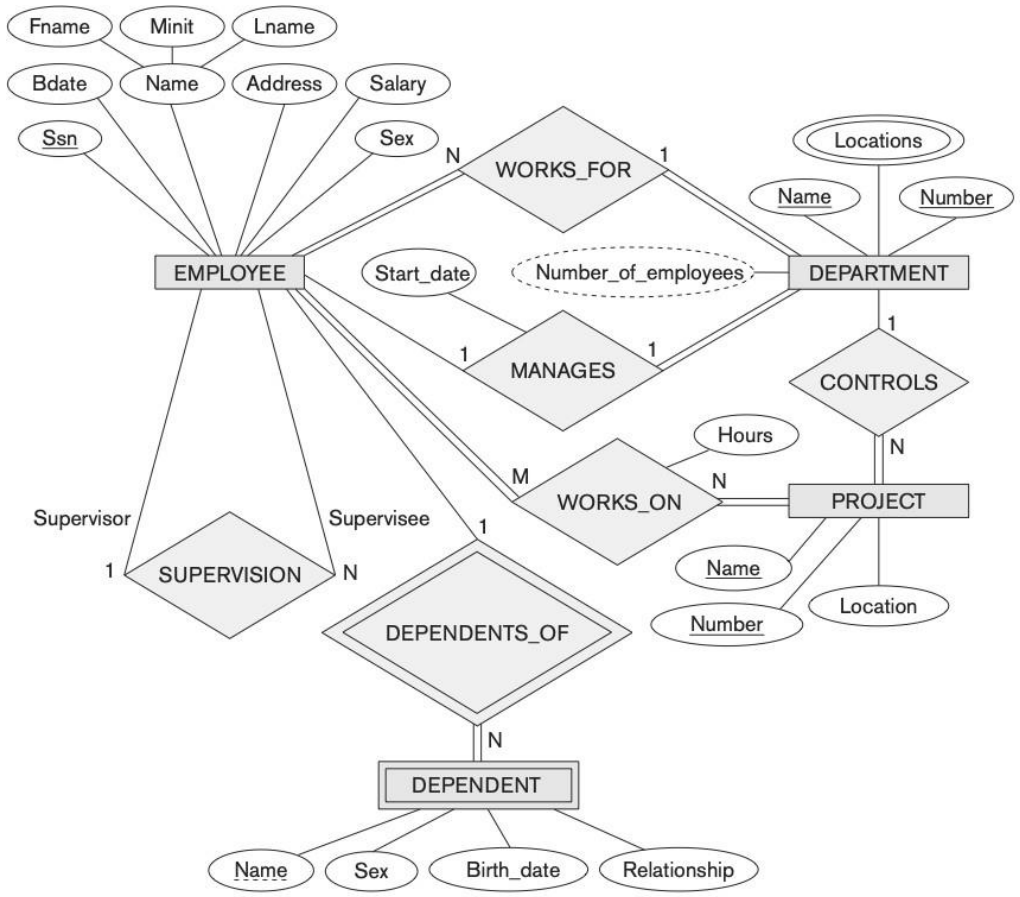

Figure 3.2
An ER schema diagram for the COMPANY database. The diagrammatic notation is introduced gradually throughout this chapter and is summarized in Figure 3.14.

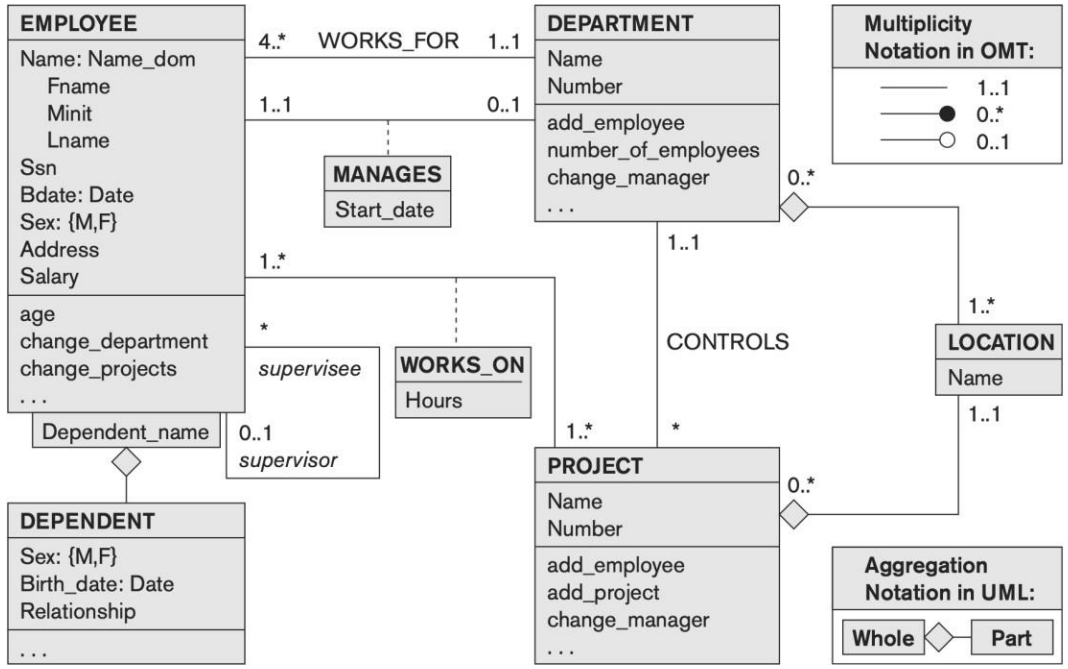

Figure 3.16
The COMPANY conceptual schema in UML class diagram notation.

Modelling Specialisation and Generalisation

- Make use of inheritance for modelling IS-A
- Extend the UML inheritance notation to make provision for overlapping constraints (filled-in triangle). Disjoint constraints makes use of the standard inheritance triangle.
- In UML classes can be modelled as Abstract, Concrete and Template. For database design the models typically make use of Concrete classes for entities.

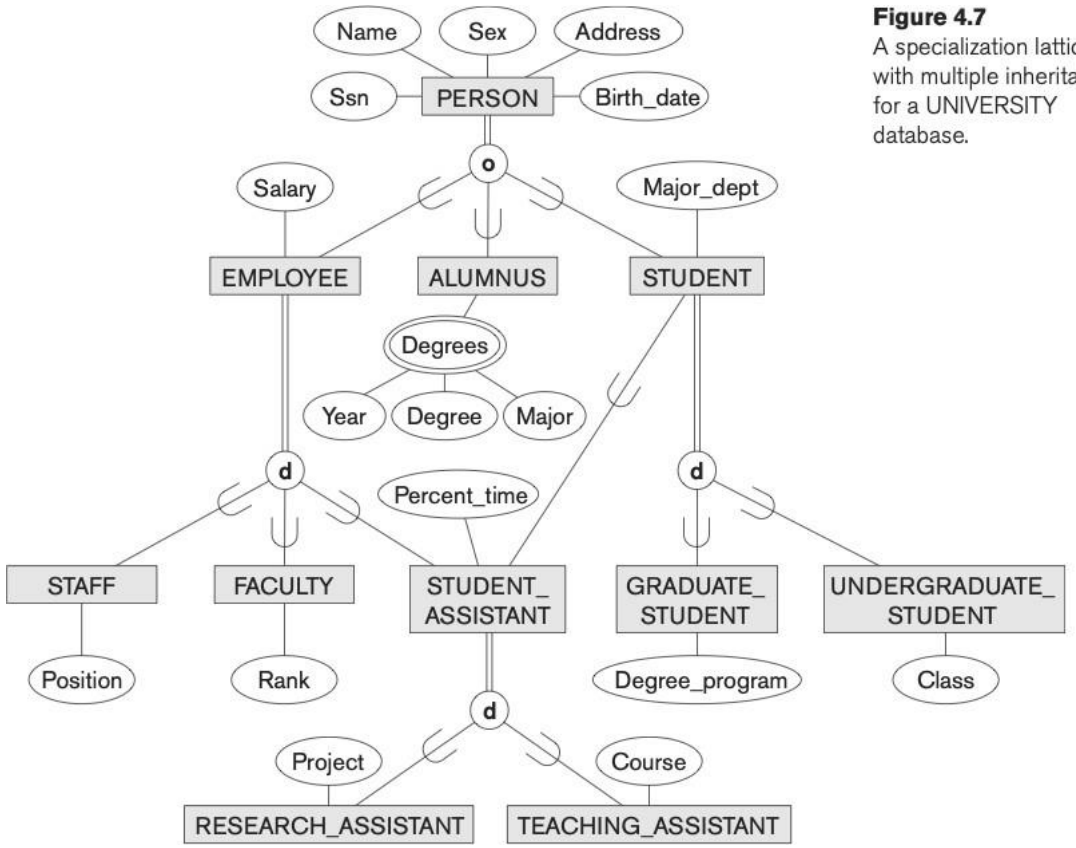

Figure 4.7
A specialization lattice with multiple inheritance for a UNIVERSITY database.

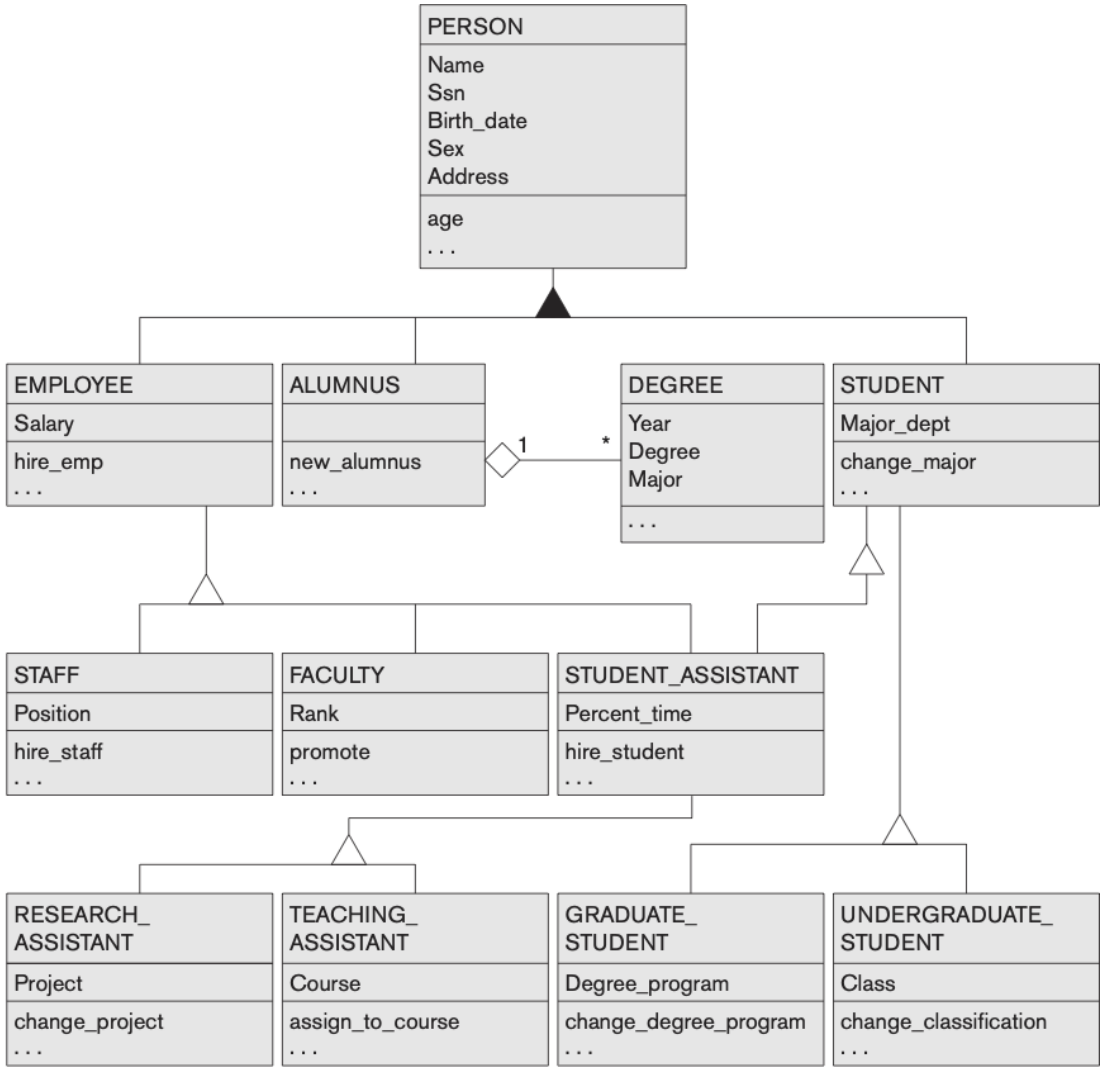

Figure 4.10
A UML class diagram corresponding to the EER diagram in Figure 4.7, illustrating UML notation for specialization/generalization.

UML Extensions

- Some tool manufacturers have developed UML extensions to model concepts such as primary keys in the UML database models. The depiction is similar to the Barker notation for primary keys.